David Khachatryan - Week 6 Report

Summary

For Milestone 1, as the graph shows, when yaw velocity is positive, the top left motor goes faster than the top right. This matches the direction of rotation that we expect. For the example below, the drone was moved manually. In figure 2, yaw input is a step function. The motor rotation remained correct.

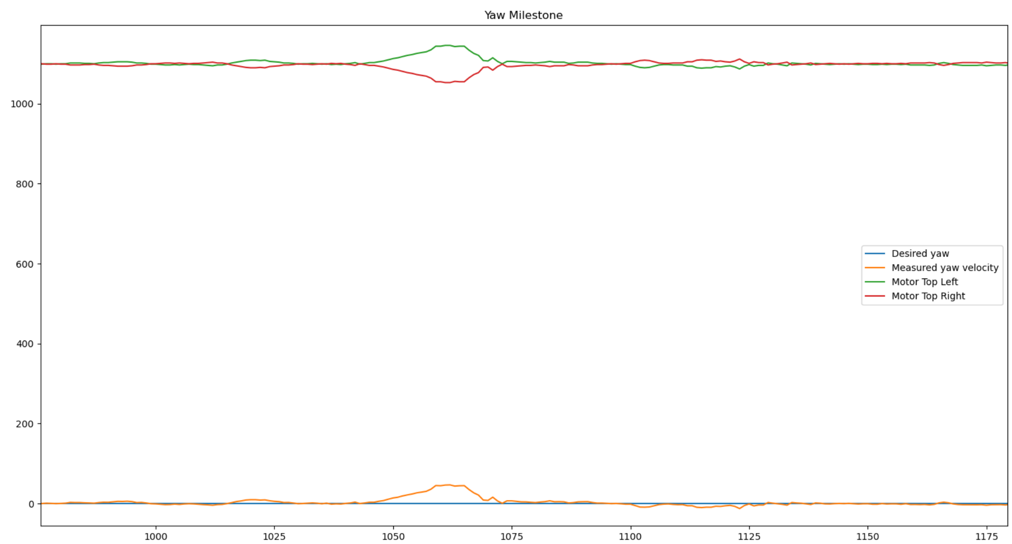

Figure 1: Measuring yaw response manual motion

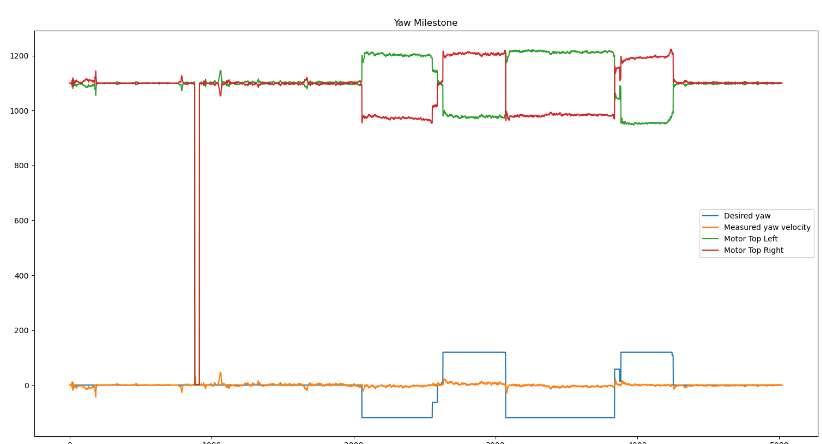

Figure 2: Showing motor responses from yaw inputs

For milestone 2, we managed to get the drone under control. It is not ideal, as it is unable to hover, but we are in control of the direction and are able to quickly adjust. The yaw took further tuning. We brought the amplitude down from the original 120 we had, which solved the issue of the drone going out of control right after takeoff.

What went well

We ran into calibration errors. Our IMU numbers suddenly became abnormal. We tackled this by running the calibration sequence in a do-while loop. It keeps running until we get within -3 to 3 range of x, y, z gyro values. This implementation saved us a lot of time we were taking to reboot the PI to fix the IMU issue.

What did not go well, why?

Our drone keeps drifting away. It has been really challenging to hover over a spot and hold position. We are not sure why, to be honest. We compartmentalized each component of RPY and they run fine individually. My assumption is that we need to continue tuning and flying.

What will you change for next class

For PID tuning, we plan to reset our numbers and follow the Ziegler–Nichols method. So far, there has not been enough intentionality in our tuning in order for us to be successful.

Contributions

We contributed equally during the class. Burke took some extra time on Monday to work on tuning without me. So I think he did 60% and I did 40%.